

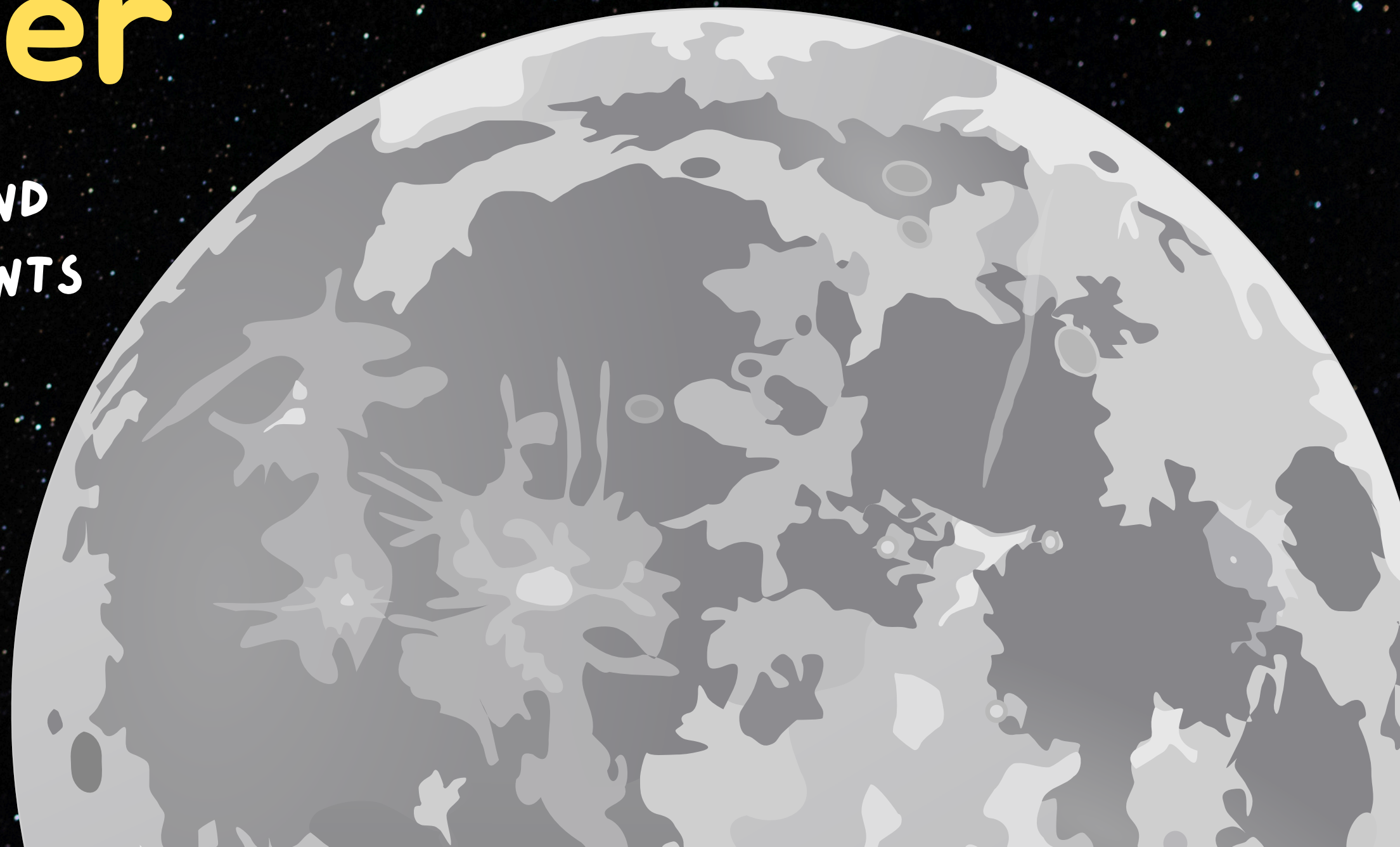
Lunar Lander

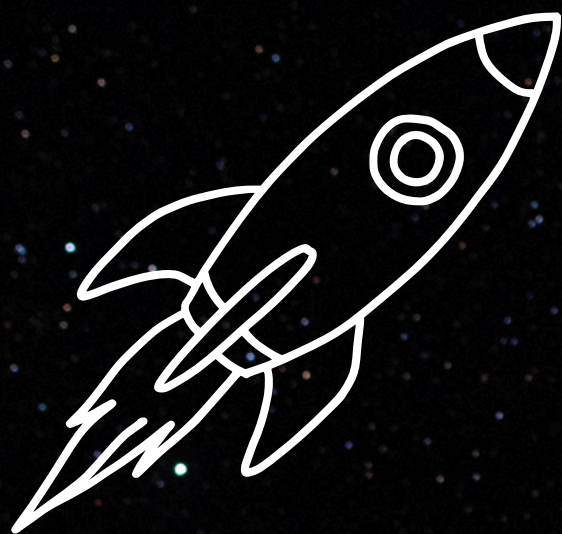
GYMNASIUM ENVIRONMENTS AND
REINFORCEMENT LEARNING AGENTS

Carolina Proença 202306055

Eduarda Neves 202307178

Maria Morais 202304201





LUNARLANDER-V3: STATES & PERCEPTS

Environment

- Physics-based 2D lunar landing task implemented with Box2D
- Fully observable continuous-state Markov Decision Process (MDP)

State / Observation Space (8D continuous vector)

At each timestep, the agent observes:

1. Horizontal position (normalized w.r.t. landing pad center)
2. Vertical position (relative to helipad height)
3. Horizontal velocity
4. Vertical velocity
5. Lander orientation angle
6. Angular velocity
7. Left leg ground contact (binary: 0 or 1)
8. Right leg ground contact (binary: 0 or 1)

Percepts

- Noise-free, real-valued observations
- Updated at 50 Hz
- State values are normalized to improve learning stability

LUNARLANDER-V3: ACTIONS & REWARDS



Action Space

Two variants (environment parameter):

- Discrete (4 actions):
 - 0: No action
 - 1: Fire left orientation engine
 - 2: Fire main engine
 - 3: Fire right orientation engine
- Continuous (2D):
 - Main engine throttle $[-1, 1]$
 - Side engine control $[-1, 1]$

Reward Function (Shaped Reward)

The reward encourages:

- Landing close to the target pad
- Low velocities at touchdown
- Upright orientation
- Stable leg contact

Reward shaping terms penalize:

- Distance from landing pad
- Linear and angular velocity
- Excessive tilt

Fuel Penalty

- Negative reward proportional to main and side engine usage

Episode Termination

- Success: lander comes to rest safely $\rightarrow +100$ reward
- Failure: crash or leaving horizontal bounds $\rightarrow -100$ reward

DEVELOPED WORK

①

Initial comparison between soft landing and horizontal descent reward formulations. Soft landing showed better learning stability and performance than horizontal descent.



②

Tested agent customization by adding extra landing legs, but no improvement was observed and the change was discarded.



③

Horizontal descent was further tested with additional penalties (Tilt penalty, Out-of-bounds penalty, Poor final positioning penalty, but it worsened the performance of horizontal descent.

⑤

Tested this environment using PPO with a CNN-based policy. Performance degraded, as the CNN relied on image-like inputs, which were less informative than the state vector.



④

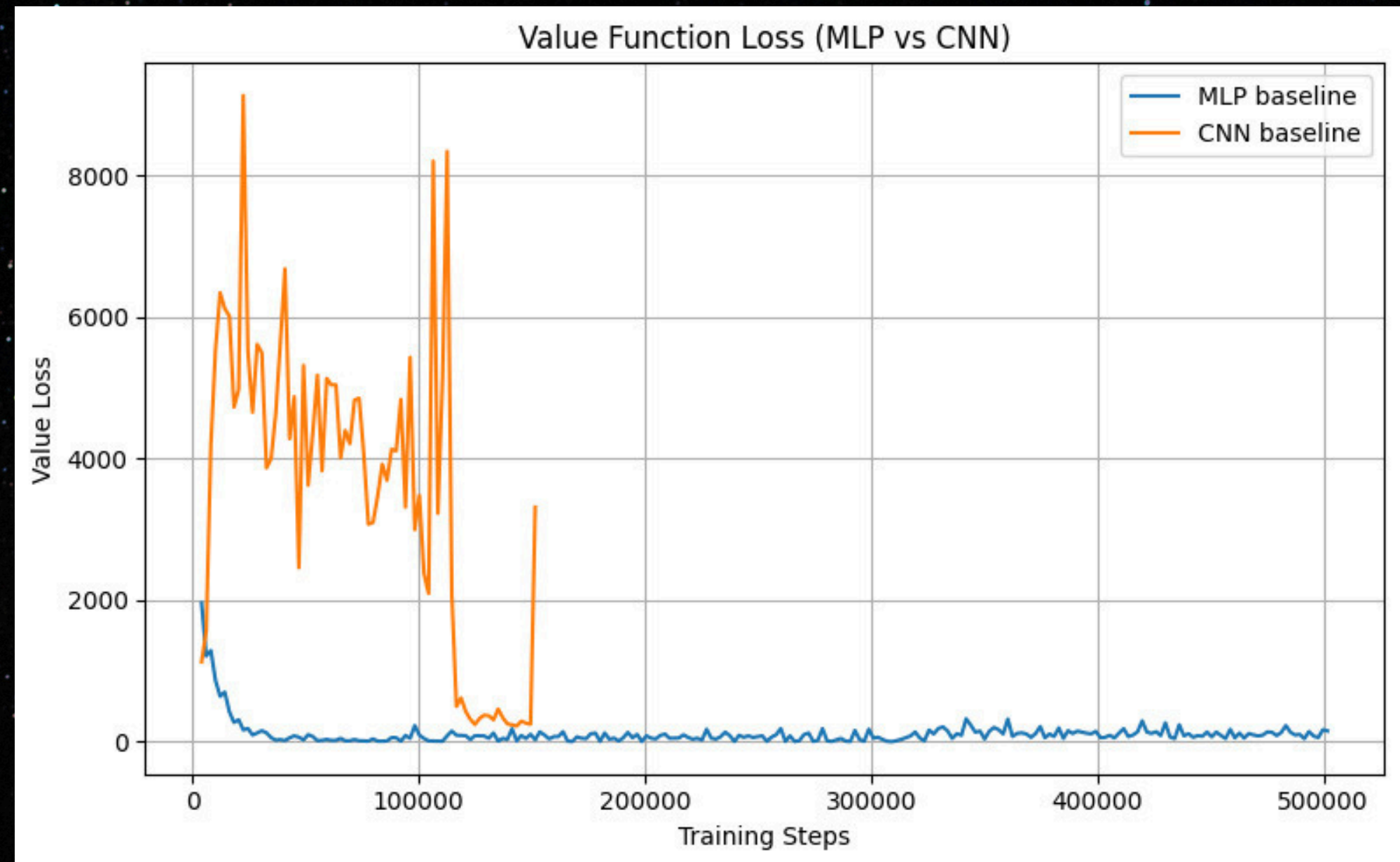
Shifted focus to soft landing combined with precision rewards and a mild tilt penalty. This soft + precision configuration achieved the best results and was fixed for subsequent experiments.



DEVELOPED WORK

COMPARISON BETWEEN MLP AND CNN TRAINING

The CNN baseline shows higher value loss compared to the MLP, indicating that its value function predictions are less accurate. This suggests that the CNN struggles more to fit the value function, leading to slower convergence during training. In contrast, the MLP exhibits lower value loss, reflecting more stable and accurate value estimates, which may contribute to more consistent learning.





DEVELOPED WORK

⑥

Tested DDPG instead of PPO, but training was very slow and unstable, proving unsuitable for this environment.



⑦

Returned to PPO + MLP on the soft + precision landing environment and increased training to 1M timesteps.



⑧

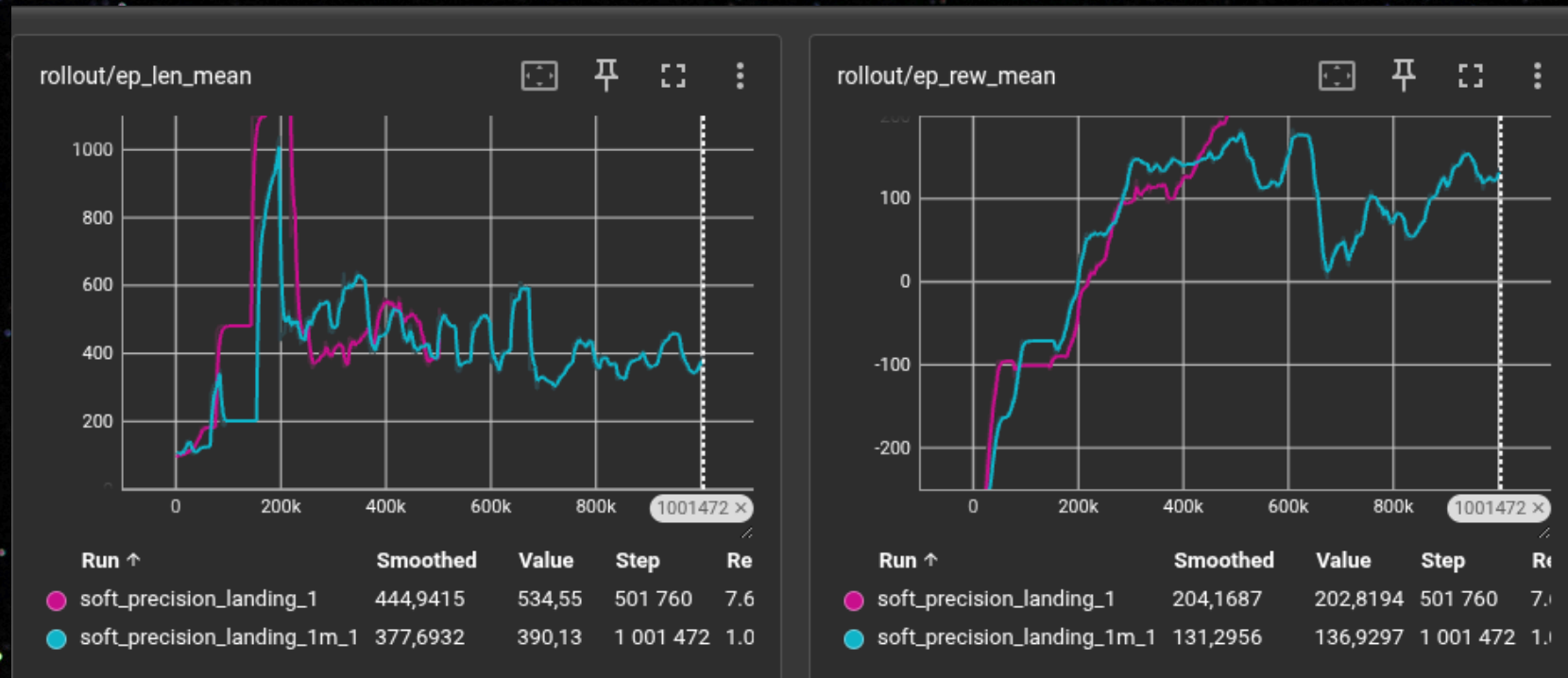
Additional reward shaping was tested to improve awareness of the helipad flags, but learning deteriorated.



⑨

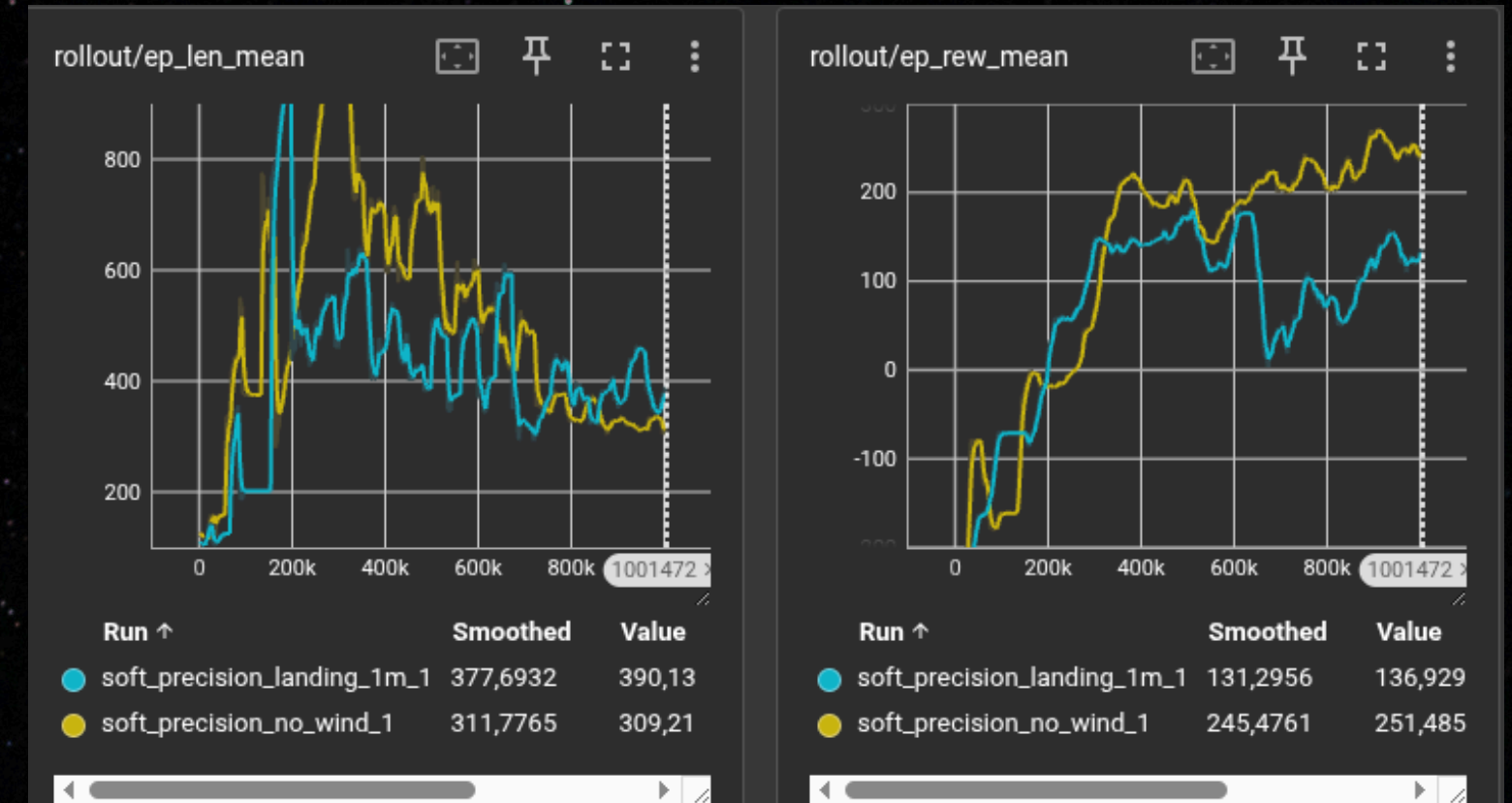
Identified wind as a major source of instability; retrained the same environment without wind, keeping 1M timesteps. Observed a significant improvement in performance, confirmed through TensorBoard metrics.

DEVELOPED WORK



COMPARISON BETWEEN 500K AND 1M TIMESTEPS

The agent trained with 1M timesteps was selected because it exhibits more stable and consistent learning, with lower variance in episode length and reward, indicating a more robust and better-converged policy than the 500k-timestep agent.



COMPARISON OF ENVIRONMENT WITH AND WITHOUT WIND

The environment without wind leads to higher average rewards and more stable episode lengths, with lower variance during training. This indicates that wind introduces significant instability, making learning harder, while training without wind results in a more robust and better-converged policy.

IMPLEMENTED CONSTRAINTS

Wind:

- Light: wind_power = 3.0, turbulence_power = 0.3
- Standard: wind_power = 15.0, turbulence_power = 3.0
- Simulates lateral disturbances that push the lander sideways, testing stability

Moving Helipad:

- Amplitude: 0.3–0.5, Frequency: 0.3–0.5
- The landing target moves horizontally to mimic a dynamic landing platform, assessing the agent's ability to track and land on a moving target.

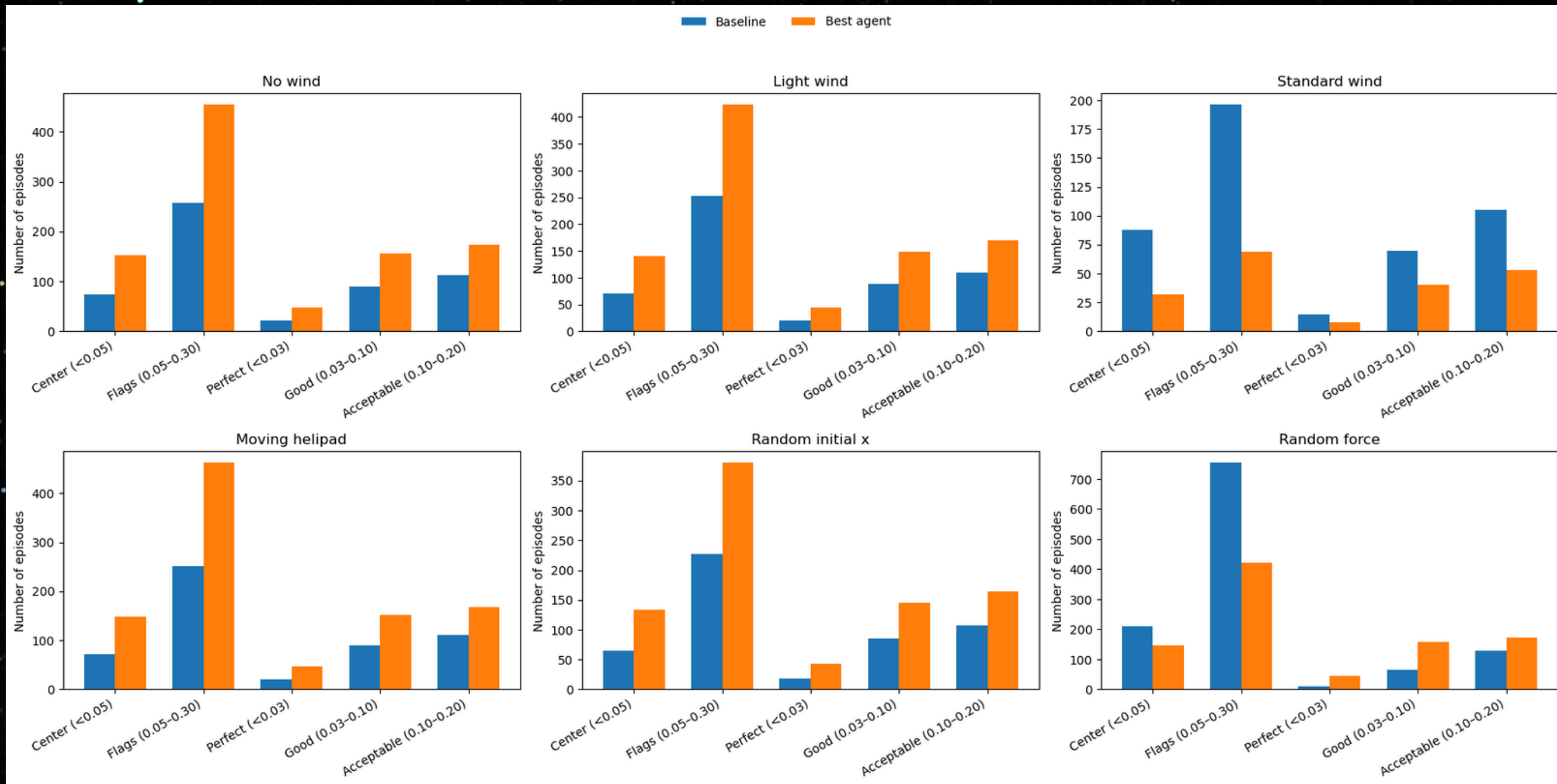
Random Initial Horizontal Position (x):

- The lander starts at a random horizontal position at the top of the environment, forcing the policy to generalize across different starting states.

Random Vertical Forces:

- Low, Medium, High magnitude downward pushes
- Tests the agent's ability to control vertical descent and maintain safe landing speeds under unexpected external forces.

COMPARISON BASELINE VS BEST AGENT



CONCLUSIONS & FUTURE WORK

- Our proposed agent consistently achieves more precise landings than baseline, especially in terms of spatial accuracy
- Landings are generally softer and more controlled → better vertical velocity management at touchdown
- Agent shows strong generalization to unseen constraints (light wind, moving helipad, random initial horizontal position and initial downforce)
- Performance degradation under strong wind highlights a clear limitation
- Overall, reward shaping successfully shifts the policy from success-only landings to precision-oriented and smooth landings

- Train with stronger wind/turbulence to improve lateral robustness
- Introduce dynamic environmental variations during training (moving helipad, random forces, wind)
- Explore policy architectures (CNN vs MLP) and hyperparameter tuning to improve value function learning and overall stability
- Investigate multi-objective optimization balancing safe impact velocity and spatial precision